

R for bioinformatics, data visualisation, part 2

HUST Bioinformatics course series

Wei-Hua Chen (CC BY-NC 4.0)

HUST, China

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Section 1

section 1: TOC

前情提要

iterations 与 并行计算

- for loop
- apply functions
- dplyr 的本质是 遍历
- map functions in purrr package
- 遍历 与 并行计算

相关包

- purrr
- parallel
- foreach
- iterators

前情提要, cont.

ggplot2 basics

- 图层 (`geom_xxx`)
- scale (`scale_xxx`)
- 坐标系统
- faceting (`facet_xxx`)

本次提要

part 2: ggplot2 进阶

- ① 如何在一张图中画多个 panel?
- ② 如何写公式
- ③ ggplot2 的核心是先计算, 再做图
- ④ themes and legends
- ⑤ stacked bars 及其它

Section 2

ggplot2 进阶 1：如何在一张图中画多个 panel?

key requirements for multi-panel plots

- order / position
- labeling
- layout

combine multiple plots

Useful packages:

- gridExtra
- cowplot
- grid
- lattice

install or load packages

```
if (!require("gridExtra")){  
  install.packages("gridExtra");  
}
```

```
if (!require("cowplot")){  
  install.packages("cowplot");  
}
```

```
library( cowplot );  
library( gridExtra );
```

arranging multiple graphs using cowplot

cowplot:

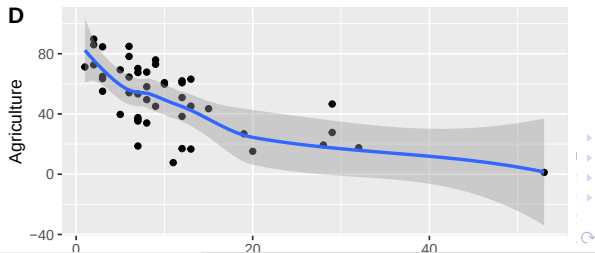
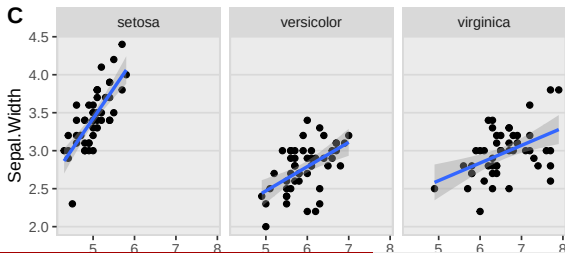
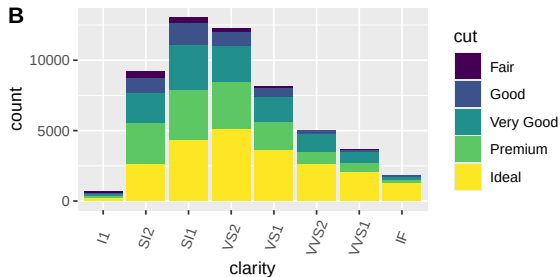
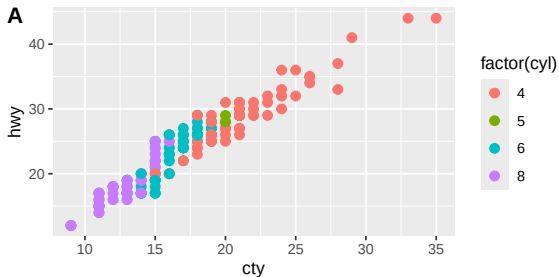
- 1 `plot_grid`
- 2 `ggdraw + draw_plot`

prepare four panels

```
library(tidyverse);
plot.hwy <- ggplot(mpg, aes(x = cty, y = hwy, colour = factor(cyl)))+
  geom_point(size=2.5)
# Bar plot
plot.dose <- ggplot(diamonds, aes(clarity, fill = cut)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle=70, vjust=0.5))
plot.iris <- ggplot(iris, aes(Sepal.Length, Sepal.Width)) +
  geom_point() + facet_grid(. ~ Species) + stat_smooth(method = "lm") +
  background_grid(major = 'y', minor = "none") + # add thin horizontal lines
  panel_border();
plot.swiss <- swiss %>% ggplot(aes(Education, Agriculture)) +
  geom_point() + geom_smooth();
```

Combine four plots using plot_grid

```
cowplot::plot_grid(plot.hwy, plot.dose, plot.iris, plot.swiss,
  labels=c("A", "B", "C", "D"), ncol = 2, nrow = 2)
```



总结，本节内容

ggplot2 基础

- 优缺点
- 用法
- 基本组成

ggplot2 进阶

- 颜色和色板
- 复杂 layout 的实现
- 公式
- ggplot2 的数据统计逻辑

更多阅读

- Ggplot2: Elegant Graphics for Data Analysis, Book by Hadley Wickham
- ggplot2 gallery provided by RStudio on Github

写在后面

- ① ggplot2 博大精深，需要一门课去讲
- ② 上手容易，精通难
- ③ 太多记忆点
- ④ 本节内容只涉及了基础中的基础，更多内容，包括进阶技巧和生信相关的扩展包，更多的需要同学们自行探索
- ⑤ 遇到不会的图，先百度/Google，找包和代码

下次预告

data summary and modeling

作业

- Exercises and homework 目录下 talk09-homework.Rmd 文件;
- 完成时间：见钉群的要求